作业

作业题目来自

梁昆淼《力学》 第 4 版 高等教育出版社

例： L01.08 梁昆淼书第 01 章第 08 题

Johns Analytical Mechanics 2nd ed Oxford University Press

例： J02.05 Johns 书第 02 章第 05 题

J03.07† 带 † 表示仅要求荣誉班同学完成

Homework 07, Week 09 2023.10.30 Due 2023.11.08 L08.04, L08.05, J05.06, J05 .09†, J15.02

★ L08.04

试应用光学中的费马原理研究光线在 x-o-y 平面上的传播，已知折射率 n 反比于坐标 y

Solution

★ L08.05

在 x-o-y 平面上给定两点，试用曲线把两点相连，使连线绕 x 轴旋转所形成的曲面面积最小

Solution

★ J05.06

A right-circular cylinder of top radius a is oriented with its symmetry axis along $\hat{\boldsymbol{e}}_3$.

(a) Use the culculus of variations and cylindrical polar coordinates to find the two differential equations describing the extremum path between two general points on the surface of the cylinder.

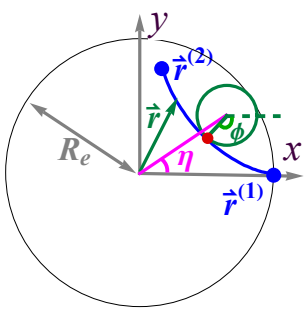
(b) Choose β equal to s, the arc length along the curve, and solve for ϕ and z as funcition of s.

(c) Suppose the cylinder surface to be cut along a line parallel to its symmetry axis, and the cut surface then flattened out onto a table. Draw a diagonal line on that flattened surface and then re-assemble the cylinder. Determing if the line you drew (now a curve, of course) is an extremum path on the surface of the cylinder.

Solution

★ J05.09 （荣誉课程同学完成）

Suppose that a rail car moves without friction through a tunnel burrowed into the earth. It starts from the rest, and moves entirely under the influence of the nonuniform gavitivity inside the earth, assumed a sphere of radius R_e with uniform density. The gavitational constant is G and mass of the earth M. Ignore the rotation of the earth. Assume the tunnel lies in the x-o-y plane, the origin of coordinates is at center of the earth, and $\hat{\boldsymbol{e}}_1$ points directly toward the point of entry.



(a) Using the general parametric methods (一般参数法），write the Euler-Lagrange equations for the path that extremizes the transit time T from the entry point to the point $\vec{r}^{(2)} = (x^{(2)}, y^{(2)}, 0)$.

(b) By choosing the parameter β as η after the Euler-Lagrange equations are written, show the solution to these equations is, for suitable choice of R, a, given by

$$x = (R-a)\cos\eta + a\cos\left(\frac{R-a}{a}\eta\right), \qquad y = (R-a)\sin\eta - a\sin\left(\frac{R-a}{a}\eta\right)$$

which are the equations of a hypocycloid, the line traced out by a point on the circumference of a circle of radius a that is rolling without slipping along the inside of a circle of radius $R > a$.

The parameter η here is the plane-polar angle of the center of the rolling circle.

(c) Suppose that the far end of the extremum tunnel is back at the surface of the earth.

If D is the distance along the surface of the earth between the entry and exit points, what is the greatest depth reached by the tunnel? How long did the trip take?

(d) Now rewrite the Euler-Lagrange equations with the choice $\beta = s$, the arc length along the unvaried path. Write a single vector equation.

本题可借助 *Mathematica* 进行一些繁琐的计算。几行代码，可省去也许是几十分钟。

Solution

★ J15.02 （诺特定理）涉及周三(11.01)的内容

Suppose that a system of N point masses has a potential function

$$U(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_N) = \sum_{n=1}^{N} \sum_{n'=1}^{n-1} f(\vec{r}_n - \vec{r}_{n'})$$

Use Noether's theorem to prove that the total momentum vector $\vec{P}$ is a constant of the motion. Is total angular momentum $\vec{J}$ also a constant of the motion?

Solution